



Turing Compatibility Guide

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Turing Compatibility Guide for CUDA Applications

The guide to building CUDA applications for NVIDIA Turing GPUs.

Chapter 1. About this Document

This application note, Turing Compatibility Guide for CUDA Applications, is intended to help developers ensure that their NVIDIA® CUDA® applications will run on GPUs based on the NVIDIA® Turing Architecture. This document provides guidance to developers who are already familiar with programming in CUDA C++ and want to make sure that their software applications are compatible with Turing.

Chapter 2. Application Compatibility on Turing

The NVIDIA CUDA C++ compiler, `nvcc`, can be used to generate both architecture-specific *cubin* files and forward-compatible *PTX* versions of each kernel. Each cubin file targets a specific compute-capability version and is forward-compatible *only with GPU architectures of the same major version number*. For example, cubin files that target compute capability 3.0 are supported on all compute-capability 3.x (Kepler) devices but are *not* supported on compute-capability 5.x (Maxwell) or 6.x (Pascal) devices. For this reason, to ensure forward compatibility with GPU architectures introduced after the application has been released, it is recommended that all applications include PTX versions of their kernels.

Note: CUDA Runtime applications containing both cubin and PTX code for a given architecture will automatically use the cubin by default, keeping the PTX path strictly for forward-compatibility purposes.

Applications that already include PTX versions of their kernels should work as-is on Turing-based GPUs. Applications that only support specific GPU architectures via cubin files, however, will need to be updated to provide Turing-compatible PTX or cubins.

Chapter 3. Compatibility between Volta and Turing

The Turing architecture is based on Volta's Instruction Set Architecture *ISA* 7.0, extending it with new instructions. As a consequence, any binary that runs on Volta will be able to run on Turing (forward compatibility), but a Turing binary will not be able to run on Volta. Please note that Volta kernels using more than 64KB of shared memory (via the explicit opt-in, see *CUDA C++ Programming Guide*) will not be able to launch on Turing, as they would exceed Turing's shared memory capacity.

Most applications compiled for Volta should run efficiently on Turing, except if the application uses heavily the Tensor Cores, or if recompiling would allow use of new Turing-specific instructions. Volta's Tensor Core instructions can only reach half of the peak performance on Turing. Recompiling explicitly for Turing is thus recommended.

Chapter 4. Verifying Turing Compatibility for Existing Applications

The first step is to check that Turing-compatible device code (at least PTX) is compiled into the application. The following sections show how to accomplish this for applications built with different CUDA Toolkit versions.

4.1. Applications Using CUDA Toolkit 8.0 or Earlier

CUDA applications built using CUDA Toolkit versions 2.1 through 8.0 are compatible with Turing as long as they are built to include PTX versions of their kernels. To test that PTX JIT is working for your application, you can do the following:

- ▶ Download and install the latest driver from <https://www.nvidia.com/drivers>.
- ▶ Set the environment variable `CUDA_FORCE_PTX_JIT=1`.
- ▶ Launch your application.

When starting a CUDA application for the first time with the above environment flag, the CUDA driver will JIT-compile the PTX for each CUDA kernel that is used into native cubin code.

If you set the environment variable above and then launch your program and it works properly, then you have successfully verified Turing compatibility.

Note: Be sure to unset the `CUDA_FORCE_PTX_JIT` environment variable when you are done testing.

4.2. Applications Using CUDA Toolkit 9.x

CUDA applications built using CUDA Toolkit 9.x are compatible with Turing as long as they are built to include kernels in either Volta-native cubin format (see [Compatibility between Volta and Turing](#)) or PTX format (see [Applications Using CUDA Toolkit 8.0 or Earlier](#)) or both.

4.3. Applications Using CUDA Toolkit 10.0

CUDA applications built using CUDA Toolkit 10.0 are compatible with Turing as long as they are built to include kernels in Volta-native or Turing-native cubin format (see [Compatibility between Volta and Turing](#)), or PTX format (see [Applications Using CUDA Toolkit 8.0 or Earlier](#)), or both.

Chapter 5. Building Applications with Turing Support

When a CUDA application launches a kernel, the CUDA Runtime determines the compute capability of each GPU in the system and uses this information to automatically find the best matching cubin or PTX version of the kernel that is available. If a cubin file supporting the architecture of the target GPU is available, it is used; otherwise, the CUDA Runtime will load the PTX and JIT-compile that PTX to the GPU's native cubin format before launching it. If neither is available, then the kernel launch will fail.

The method used to build your application with either native cubin or at least PTX support for Turing depend on the version of the CUDA Toolkit used.

The main advantages of providing native cubins are as follows:

- ▶ It saves the end user the time it takes to JIT-compile kernels that are available only as PTX. All kernels compiled into the application must have native binaries at load time or else they will be built just-in-time from PTX, including kernels from all libraries linked to the application, even if those kernels are never launched by the application. Especially when using large libraries, this JIT compilation can take a significant amount of time. The CUDA driver will cache the cubins generated as a result of the PTX JIT, so this is mostly a one-time cost for a given user, but it is time best avoided whenever possible.
- ▶ PTX JIT-compiled kernels often cannot take advantage of architectural features of newer GPUs, meaning that native-compiled code may be faster or of greater accuracy.

5.1. Applications Using CUDA Toolkit 8.0 or Earlier

The compilers included in CUDA Toolkit 8.0 or earlier generate cubin files native to earlier NVIDIA architectures such as Maxwell and Pascal, but they *cannot* generate cubin files native to Volta or Turing architecture. To allow support for Volta, Turing and future architectures when using version 8.0 or earlier of the CUDA Toolkit, the compiler must generate a PTX version of each kernel.

Below are compiler settings that could be used to build `mykernel.cu` to run on Maxwell or Pascal devices natively and on Turing devices via PTX JIT.

Note: `compute_XX` refers to a PTX version and `sm_XX` refers to a cubin version. The `arch=` clause of the `-gencode=` command-line option to `nvcc` specifies the front-end compilation target and must always be a PTX version. The `code=` clause specifies the back-end compilation target and can either

be cubin or PTX or both. Only the back-end target version(s) specified by the `code=` clause will be retained in the resulting binary; at least one must be PTX to provide Turing compatibility.

Windows

```
nvcc.exe -ccbin "C:\vs2010\VC\bin"
-Xcompiler "/EHsc /W3 /nologo /O2 /Zi /MT"
-gencode=arch=compute_50,code=sm_50
-gencode=arch=compute_52,code=sm_52
-gencode=arch=compute_60,code=sm_60
-gencode=arch=compute_61,code=sm_61
-gencode=arch=compute_61,code=compute_61
--compile -o "Release\mykernel.cu.obj" "mykernel.cu"
```

Mac/Linux

```
/usr/local/cuda/bin/nvcc
-gencode=arch=compute_50,code=sm_50
-gencode=arch=compute_52,code=sm_52
-gencode=arch=compute_60,code=sm_60
-gencode=arch=compute_61,code=sm_61
-gencode=arch=compute_61,code=compute_61
-O2 -o mykernel.o -c mykernel.cu
```

Alternatively, you may be familiar with the simplified `nvcc` command-line option `-arch=sm_XX`, which is a shorthand equivalent to the following more explicit `-gencode=` command-line options used above. `-arch=sm_XX` expands to the following:

```
-gencode=arch=compute_XX,code=sm_XX
-gencode=arch=compute_XX,code=compute_XX
```

However, while the `-arch=sm_XX` command-line option does result in inclusion of a PTX back-end target by default, it can only specify a single target cubin architecture at a time, and it is not possible to use multiple `-arch=` options on the same `nvcc` command line, which is why the examples above use `-gencode=` explicitly.

5.2. Applications Using CUDA Toolkit 9.x

With versions 9.x of the CUDA Toolkit, `nvcc` can generate cubin files native to the Volta architecture (compute capability 7.0). When using CUDA Toolkit 9.x, to ensure that `nvcc` will generate cubin files for all recent GPU architectures as well as a PTX version for forward compatibility with future GPU architectures, specify the appropriate `-gencode=` parameters on the `nvcc` command line as shown in the examples below.

Windows

```
nvcc.exe -ccbin "C:\vs2010\VC\bin"
-Xcompiler "/EHsc /W3 /nologo /O2 /Zi /MT"
-gencode=arch=compute_50,code=sm_50
-gencode=arch=compute_52,code=sm_52
-gencode=arch=compute_60,code=sm_60
-gencode=arch=compute_61,code=sm_61
-gencode=arch=compute_70,code=sm_70
```

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```
-gencode=arch=compute_70,code=compute_70
--compile -o "Release\mykernel.cu.obj" "mykernel.cu"
```

Mac/Linux

```
/usr/local/cuda/bin/nvcc
-gencode=arch=compute_50,code=sm_50
-gencode=arch=compute_52,code=sm_52
-gencode=arch=compute_60,code=sm_60
-gencode=arch=compute_61,code=sm_61
-gencode=arch=compute_70,code=sm_70
-gencode=arch=compute_70,code=compute_70
-O2 -o mykernel.o -c mykernel.cu
```

Note: compute_XX refers to a PTX version and sm_XX refers to a cubin version. The arch= clause of the -gencode= command-line option to nvcc specifies the front-end compilation target and must always be a PTX version. The code= clause specifies the back-end compilation target and can either be cubin or PTX or both. Only the back-end target version(s) specified by the code= clause will be retained in the resulting binary; at least one should be PTX to provide compatibility with future architectures.

Also, note that CUDA 9.0 removes support for compute capability 2.x (Fermi) devices. Any compute_2x and sm_2x flags need to be removed from your compiler commands.

5.3. Applications Using CUDA Toolkit 10.0

With version 10.0 of the CUDA Toolkit, nvcc can generate cubin files native to the Turing architecture (compute capability 7.5). When using CUDA Toolkit 10.0, to ensure that nvcc will generate cubin files for all recent GPU architectures as well as a PTX version for forward compatibility with future GPU architectures, specify the appropriate -gencode= parameters on the nvcc command line as shown in the examples below.

Windows

```
nvcc.exe -ccbin "C:\vs2010\VC\bin"
-Xcompiler "/EHsc /W3 /nologo /O2 /Zi /MT"
-gencode=arch=compute_50,code=sm_50
-gencode=arch=compute_52,code=sm_52
-gencode=arch=compute_60,code=sm_60
-gencode=arch=compute_61,code=sm_61
-gencode=arch=compute_70,code=sm_70
-gencode=arch=compute_75,code=sm_75
-gencode=arch=compute_75,code=compute_75
--compile -o "Release\mykernel.cu.obj" "mykernel.cu"
```

Mac/Linux

```
/usr/local/cuda/bin/nvcc
-gencode=arch=compute_50,code=sm_50
-gencode=arch=compute_52,code=sm_52
-gencode=arch=compute_60,code=sm_60
-gencode=arch=compute_61,code=sm_61
```

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```
-gencode=arch=compute_70,code=sm_70  
-gencode=arch=compute_75,code=sm_75  
-gencode=arch=compute_75,code=compute_75  
-O2 -o mykernel.o -c mykernel.cu
```

Note: `compute_XX` refers to a PTX version and `sm_XX` refers to a cubin version. The `arch=` clause of the `-gencode=` command-line option to `nvcc` specifies the front-end compilation target and must always be a PTX version. The `code=` clause specifies the back-end compilation target and can either be cubin or PTX or both. Only the back-end target version(s) specified by the `code=` clause will be retained in the resulting binary; at least one should be PTX to provide compatibility with future architectures.

5.4. Independent Thread Scheduling Compatibility

The Volta and Turing architectures feature Independent Thread Scheduling among threads in a warp. If the developer made assumptions about warp-synchronicity,¹ this feature can alter the set of threads participating in the executed code compared to previous architectures. Please see Compute Capability 7.0 in the *CUDA C++ Programming Guide* for details and corrective actions. To aid migration Volta and Turing developers can opt-in to the Pascal scheduling model with the following combination of compiler options.

```
nvcc -arch=compute_60 -code=sm_70 ...
```

¹ *Warp-synchronous* refers to an assumption that threads in the same warp are synchronized at every instruction and can, for example, communicate values without explicit synchronization.

Chapter 6. Revision History

Version 1.0

- ▶ Initial public release.

Version 1.1

- ▶ Use CUDA C++ instead of CUDA C/C++
- ▶ Updated references to the CUDA C++ Programming Guide and CUDA C++ Best Practices Guide.

Chapter 7. Notices

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